Ali El Seblani

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Education

Harvey Mudd College Claremont, CA
Bachelor of Science, Engineering | GPA: 3.89 May 2028

Relevant Coursework: Mechanical Vibration, Experimental Engineering, Signals and Systems, Thermodynamics, Applied Math for Engineering

Skills

CAD & Simulation: SolidWorks, Onshape, Fusion 360, COMSOL, FEA, Modal Analysis, CFD
Design & Manufacturing: GD&T, DFM, DFA, LSS, Design for Six Sigma (Green Belt)
Fabrication: 2-Axis Machining, CNC Machining, Additive Manufacturing, Laser & Waterjet Cutting, MIG Welding
Programming: MATLAB, Python, C++

Engineering Experience

General Motors Burton, MI
Autonomous Robotics Center Manufacturing Engineering Intern May 2026 – August 2026

- Reengineered material-handling processes across 4 departments supporting 150+ operators to integrate an Automated Storage and Retrieval System, autonomous mobile robots, and conveyor systems, reducing material touches by 63%
- Analyzed production volumes across 5,000+ part numbers and formulated 5 workstation concepts to define capacity requirements and evaluate throughput, footprint, ergonomics, safety, sustainability, and implementation cost
- Mapped current- and future-state process-flow models using standard-time data to identify bottlenecks and shorten processing time by 12%
- Modeled simulated work environments and applied Design for Six Sigma methodologies to validate future-state processes, decreasing operator motion by 36%, increasing throughput capacity by 13%, and supporting a processing strategy projected to save $1M annually

Harvey Mudd Formula SAE Claremont, CA
Powertrain Lead December 2025 – Present

- Leading a 7+ member Powertrain team through design, analysis, and integration of engine systems (cooling and fuel), transmission and shifting mechanisms, and differential assemblies, conducting weekly design reviews to align subsystem timelines and technical decisions
- Designed cooling and fuel systems, an eccentric chain-tensioning mechanism, and differential mounts; used Onshape Simulation for linear static and modal analysis to evaluate stress, factor of safety, and natural frequencies, reducing component mass through multiple design iterations while meeting strength, stiffness, and vibration requirements
- Selecting final drive reduction based on the Honda CBR600F4i transmission ratios, engine torque/RPM curve, tire diameter, and target acceleration performance, then designing the mechanical shifting linkage and validating actuation reliability through bench testing
- Coordinating with chassis, suspension, and electrical team to resolve powertrain packaging and interface conflicts while ensuring component designs meet manufacturing and serviceability requirements

Harvey Mudd College Claremont, CA
Systems Engineering Practicum Proctor May 2026 – Present

- Lead weekly Systems Engineering practicums, teaching fundamental systems and signals concepts through hands-on experiments
- Guide 30+ students in constructing and testing an underwater robot incorporating proportional control, analog sensing circuits, and standard lab instrumentation to reinforce system integration and experimental practice

Machine Shop Proctor December 2025 – Present

- Supervise safe operation of a facility with 20+ manual, CNC, and fabrication machines for 80+ engineering students each semester
- Guide 80+ students in interpreting engineering drawings and process routers, selecting machining operations, and manufacturing components to dimensional specifications

Project Experience

Autonomous Ocean Profiling Robot for Climate Monitoring | *Engineering Project* March 2026 – May 2026

- Engineered and assembled an autonomous underwater robot to collect pressure, temperature, and ultraviolet measurements at depths up to 3 meters during marine field testing
- Led electrical-system development for ultraviolet, pressure, and temperature sensors, implementing unity-gain buffer and inverting-amplifier circuits to condition sensor outputs to 0.3–3 V, utilizing 82% of the Teensy 4.0 ADC range and preventing signal clipping
- Developed sensor-calibration procedures accounting for noise, sensor nonlinearity, and degradation, matching ground-truth temperature within 0.6 degrees Celsius across laboratory and field conditions
- Developed a C++ PID depth-control algorithm to reach and maintain underwater waypoints and built MATLAB scripts showing UV intensity falling 84% to 3 meters while temperature held near 18.1 degrees Celsius

Machinist's Hammer | *Engineering Project* September 2025 – October 2025

- Calibrated and operated manual mills and lathes to manufacture a machinist's hammer from raw steel and nylon to thousandths-of-an-inch specifications
- Performed drilling, tapping, heat treatment, and tempering to achieve specified dimensions and material properties